

3

Non-linear Partial Differential Equations of Order One

3.1 Explanation of terms: Complete integral (or complete solution), particular integral, singular integral (or singular solution), and general integral (or general solution) as applied to solutions of first order partial differential equations

A *solution* or *integral* of a differential equation is a relation between the variables, by means of which and the derivatives obtained there from the equation is satisfied. Let us now discuss various classes of integrals of a partial differential equation of order one.

Complete Integral (C. I.) or complete solution (C.S.)

Let us consider a relation $\phi(x, y, z, a, b) = 0$... (1)
in which x, y, z are variables such that z is dependent on x and y . Differentiating (1) partially w.r.t x and y respectively, we obtain

$$\frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial z} p = 0 \quad \text{and} \quad \frac{\partial \phi}{\partial y} + \frac{\partial \phi}{\partial z} q = 0, \quad \dots (2)$$

Since there are two arbitrary constants (namely a and b) connected by the above three equations, these can be eliminated and there will appear a relation of the form

$$f(x, y, z, p, q) = 0, \quad \dots (3)$$

which is a partial differential equation of order one.

Suppose, now, that (1) has been derived from (3), by using some method; then the integral (1), which has as many arbitrary constants as there are independent variables, is called the *complete integral* of (3).

Particular Integral : A particular integral of (3) is obtained by giving particular values to a and b in (1) which is the complete integral of (3).

Singular Integral (S.I.) or singular solution (S.S.)

[Delhi 2009]

We know that the locus of all the points whose co-ordinates along with the values of p and q satisfy (3), represent the doubly infinite system of surfaces given by (1). The system is doubly infinite, since there are two constants a and b and each of these can take an infinite number of values. Since the envelope of all the surfaces given by (1) is touched at each of its points by some one of these surfaces, the coordinates of any point on the envelope along with the values of p and q belonging to the envelope at that point must also satisfy (3). Hence we conclude that the equation of the envelope is a solution of (3). The envelope of the surfaces given by (2) is obtained by eliminating a and b between the equations

$$\phi(x, y, z, a, b) = 0, \quad \frac{\partial \phi}{\partial a} = 0 \quad \text{and} \quad \frac{\partial \phi}{\partial b} = 0. \quad \dots (4)$$

The relation between x, y , and z so obtained is called the *singular integral*. In general, it is distinct from the complete integral. However, in exceptional cases it may be contained in the

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complete integral, that is, singular integral may be obtained by giving particular values to the constants in the complete integral. Since other relations may appear in the process of getting the singular integral, it is necessary to test that the equation of singular integral satisfies the given differential equation.

General Integral (G.I.) or General Solution (G.S.).

Assume that in (1), one of the constants is a function of the other, say $b = F(a)$, then (1) becomes

$$\phi(x, y, z, a, F(a)) = 0. \quad \dots(5)$$

Now (5) represents one of the families of surfaces given by the system (1). As before, the equation of the envelope of the family of surfaces given by (5) must also satisfy (3). Again the equation so obtained will be distinct from that of the envelope of the surfaces, and it is not a particular integral. It is known as the *general integral* and is obtained by eliminating a between

$$\phi(x, y, z, a, F(a)) = 0 \quad \text{and} \quad \partial\phi/\partial a = 0. \quad \dots(6)$$

Since other relations may appear in the process of getting the singular integral, it is necessary to test that the equation of general integral satisfies the given differential equation.

Important Note. While solving a non-linear equation, we must not only obtain the complete integral but should also find the singular and general integrals. In absence of details of singular and general integrals, merely the complete solution is considered to be incomplete solution of the given partial differential equation. However, for reason of space, we have found complete integral only in some problems. The students are advised to find singular and general integrals also for such problems. Note that there is always a simple routine method for the same.

Also, if you are asked to find complete integral of a given equation, then you need not give singular and general integrals. Again, if examiner wants singular integral/general integral, then you must find them.

3.2 Geometrical interpretation of three types of integrals of $f(x, y, z, p, q) = 0$.

(i) Complete integral.

A complete integral, being a relation between x, y and z represents equation of a surface. Since it involves two arbitrary parameters, it belongs to a double infinite system of surfaces or to a single infinite system of family of surfaces.

(ii) General integral.

Let a complete solution of $f(x, y, z, p, q) = 0$ be

$$\phi(x, y, z, a, b) = 0. \quad \dots(1)$$

A general integral is obtained by eliminating ' a ' between (1) and the equations

$$b = \psi(a) \quad \dots(2)$$

$$(\partial\phi/\partial a) + (\partial\phi/\partial b) \psi'(a) = 0. \quad \dots(3)$$

where ψ is an arbitrary function.

The operation of elimination is equivalent to selecting from the system of families of surfaces a representative family and finding the envelope. Equations (1), (2) and (3) together represent a curve drawn on the surface of the family whose parameter is ' a ' whereas the equation obtained by eliminating ' a ' between them is the envelope of the family. It follows that the envelope touches the surface represented by (1) and (2) along the curve represented by (1), (2) and (3). This curve is known as *characteristic of the envelope* and the general integral thus represents the envelope of a family of surfaces considered as composed of its characteristics.

(iii) Singular Integral.

The singular integral is obtained by eliminating ' a ' and ' b ' between equation (1)

$$\partial\phi/\partial a = 0 \quad \dots(4)$$

and $\partial\phi/\partial b = 0. \quad \dots(5)$

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Integrating, $2 \log z = \log x + \log y + \log C_1$ or $z^2 = C_1 xy \dots (6)$

Similarly, using the corresponding values $p = xz / (x^2 + y^2)$ and $q = yz / (x^2 + y^2)$ in $dz = p dx + q dy$, we get

$$dz = \frac{xz dx}{x^2 + y^2} + \frac{yz dy}{x^2 + y^2} \quad \text{or} \quad \frac{2dz}{z} = \frac{2(xdx + ydy)}{x^2 + y^2}$$

Integrating, $2 \log z = \log(x^2 + y^2) + \log C_2$ or $z^2 = C_2(x^2 + y^2) \dots (7)$

(6) and (7) give two common solutions of (1) and (2)

EXERCISE 3(A)

1. Show that $\partial z / \partial x = 7x + 8y - 1$ and $\partial z / \partial y = 9x + 11y - 2$ are not compatible.
2. Show that the partial differential equations $p = 6x - 4y + 1$ and $q = 4x + 6y + 1$ do not possess any common solution.
3. Show that the following system of partial differential equations are compatible and hence solve them

- | | |
|--|---|
| (i) $p = 6x + 3y, q = 3x - 4y$ | Ans. $z = 3x^2 + 3xy - 2y^2 + c$ |
| (ii) $p = ax + hy + g, q = hx + by + f$ | Ans. $z = (ax^2 + by^2)/2 + hxy + gx + fy + c$ |
| (iii) $\partial z / \partial x = y(2ax + by), \partial z / \partial y = x(ax + 2by)$ | Ans. $z = ax^2y + bxy^2 + c$ |
| (iv) $p = x^4 - 2xy^2 + y^4, q = 4xy^3 - 2x^2y - \sin y$ | Ans. $z = x^5/5 - x^2y^2 + xy^4 + \cos y + c$ |
| (v) $p = (e^y + 1) \cos x, q = e^y \sin x$ | Ans. $z = (e^y + 1) \sin x + c$ |
| (vi) $p = y(1 + 1/x) + \cos y, q = x + \log x - x \sin y$ | Ans. $z = y(x + \log x) + x \cos y + c$ |
| (vii) $p = y^2 e^{xy^2} + 4x^3, q = 2xy e^{xy^2} - 3y^2$ | Ans. $z = e^{xy^2} + x^4 - y^3 + c$ |
| (viii) $p = \sin x \cos y + e^{3x}, q = \cos x \sin y + \tan y$ | Ans. $z = (1/3) \times e^{3x} - \cos x \cos y + \log \sec x + c$ |

3.7 Charpit's method.* (General method of solving partial differential equations of order one but of any degree)

[Agra 2003; Delhi B.Sc. (Hons.) 2011,

Avadh 2006; Lucknow 2011; Meerut 2005; Nagpur 2008; Rohilkhand 2004]

Let the given partial equation differential of first order and non-linear in p and q be

$$f(x, y, z, p, q) = 0. \quad \dots (1)$$

We know that

$$dz = p dx + q dy. \quad \dots (2)$$

The next step consists in finding another relation $F(x, y, z, p, q) = 0$

$$\dots (3)$$

such that when the values of p and q obtained by solving (1) and (3), are substituted in (2), it becomes integrable. The integration of (2) will give the complete integral of (1).

In order to obtain (3), differentiate partially (1) and (3) with respect to x and y and get

$$\frac{\partial f}{\partial x} + \frac{\partial f}{\partial z} p + \frac{\partial f}{\partial p} \frac{\partial p}{\partial x} + \frac{\partial f}{\partial q} \frac{\partial q}{\partial x} = 0, \quad \dots (4)$$

$$\frac{\partial F}{\partial x} + \frac{\partial F}{\partial z} p + \frac{\partial F}{\partial p} \frac{\partial p}{\partial x} + \frac{\partial F}{\partial q} \frac{\partial q}{\partial x} = 0, \quad \dots (5)$$

$$\frac{\partial f}{\partial y} + \frac{\partial f}{\partial z} q + \frac{\partial f}{\partial p} \frac{\partial p}{\partial y} + \frac{\partial f}{\partial q} \frac{\partial q}{\partial y} = 0 \quad \dots (6)$$

*This is general method for solving equations with two independent variables. Since the solution by this method is generally more complicated, this method is applied to solve equations which cannot be reduced to any of the standard forms which will be discussed later on. Thus, Charpit's method is used in two situations

(i) When you are asked to solve a problem by Charpit's method (ii) when the given equation is not of any four standard forms given in Articles 3.10, 3.12, 3.14 and 3.17.

and
$$\frac{\partial F}{\partial y} + \frac{\partial F}{\partial z} q + \frac{\partial F}{\partial p} \frac{\partial p}{\partial y} + \frac{\partial F}{\partial q} \frac{\partial q}{\partial y} = 0. \quad \dots (7)$$

Eliminating $\partial p / \partial x$ from (4) and (5), we get

$$\left(\frac{\partial f}{\partial x} + \frac{\partial f}{\partial z} p + \frac{\partial f}{\partial q} \frac{\partial q}{\partial x} \right) \frac{\partial F}{\partial p} - \left(\frac{\partial F}{\partial x} + \frac{\partial F}{\partial z} p + \frac{\partial F}{\partial q} \frac{\partial q}{\partial x} \right) \frac{\partial f}{\partial p} = 0$$

or
$$\left(\frac{\partial f}{\partial x} \frac{\partial F}{\partial p} - \frac{\partial F}{\partial x} \frac{\partial f}{\partial p} \right) + \left(\frac{\partial f}{\partial z} \frac{\partial F}{\partial p} - \frac{\partial F}{\partial z} \frac{\partial f}{\partial p} \right) p + \left(\frac{\partial f}{\partial q} \frac{\partial F}{\partial p} - \frac{\partial F}{\partial q} \frac{\partial f}{\partial p} \right) \frac{\partial q}{\partial x} = 0. \quad \dots (8)$$

Similarly, eliminating $\partial q / \partial y$ from (6) and (7), we get

$$\left(\frac{\partial f}{\partial y} \frac{\partial F}{\partial q} - \frac{\partial F}{\partial y} \frac{\partial f}{\partial q} \right) + \left(\frac{\partial f}{\partial z} \frac{\partial F}{\partial q} - \frac{\partial F}{\partial z} \frac{\partial f}{\partial q} \right) q + \left(\frac{\partial f}{\partial p} \frac{\partial F}{\partial q} - \frac{\partial F}{\partial p} \frac{\partial f}{\partial q} \right) \frac{\partial p}{\partial y} = 0. \quad \dots (9)$$

Since $\partial q / \partial x = \partial^2 z / \partial x \partial y = \partial p / \partial y$, the last term in (8) is the same as that in (9), except for a minus sign and hence they cancel on adding (8) and (9).

Therefore, adding (8) and (9) and rearranging the terms, we obtain

$$\left(\frac{\partial f}{\partial x} + p \frac{\partial f}{\partial z} \right) \frac{\partial F}{\partial p} + \left(\frac{\partial f}{\partial y} + \frac{\partial f}{\partial z} q \right) \frac{\partial F}{\partial q} + \left(-p \frac{\partial f}{\partial p} - q \frac{\partial f}{\partial q} \right) \frac{\partial F}{\partial z} + \left(-\frac{\partial f}{\partial p} \right) \frac{\partial F}{\partial x} + \left(-\frac{\partial f}{\partial q} \right) \frac{\partial F}{\partial y} = 0. \quad \dots (10)$$

This is a linear equation of the first order to obtain the desired function F . As in Art 2.20 of chapter 2, integral of (10) is obtained by solving the auxiliary equations

$$\frac{dp}{(\partial f / \partial x) + p(\partial f / \partial z)} = \frac{dq}{(\partial f / \partial y) + q(\partial f / \partial z)} = \frac{dz}{-p(\partial f / \partial p) - q(\partial f / \partial q)} = \frac{dx}{-\partial f / \partial p} = \frac{dy}{-\partial f / \partial q} = \frac{dF}{0}. \quad \dots (11)$$

Since any of the integrals of (11) will satisfy (10), an integral of (11) which involves p or q (or both) will serve along with the given equation to find p and q . In practice, however, we shall select the simplest integral.

Note. In what follows we shall use the following standard notations:

$$\partial f / \partial x = f_x, \quad \partial f / \partial y = f_y, \quad \partial f / \partial z = f_z, \quad \partial f / \partial p = f_p, \quad \partial f / \partial q = f_q.$$

Therefore, Charpit's auxiliary equations (11) may be re-written as

$$\frac{dp}{f_x + p f_z} = \frac{dq}{f_y + q f_z} = \frac{dz}{-p f_p - q f_q} = \frac{dx}{-f_p} = \frac{dy}{-f_q} = \frac{dF}{0} \quad \dots (11)'$$

3.8a Working rule while using charpit's method

Step 1. Transfer all terms of the given equation to L.H.S. and denote the entire expression by f .

Step 2. Write down the Charpit's auxiliary equations (11) or (11)′.

Step 3. Using the value of f in step 1 write down the values of $\partial f / \partial x, \partial f / \partial y, \dots$, i.e., $f_x, f_y, \dots$ etc. occurring in step 2 and put these in Charpit's equations (11) or (11)′.

Step 4. After simplifying the step 3, select two proper fractions so that the resulting integral may come out to be the simplest relation involving at least one of p and q .

Step 5. The simplest relation of step 4 is solved along with the given equation to determine p and q . Put these values of p and q in $dz = p dx + q dy$ which on integration gives the complete integral of the given equation.

The Singular and General integrals may be obtained in the usual manner.

Remark. Sometimes Charpit's equations give rise to $p = a$ and $q = b$, where a and b are constants. In such cases, putting $p = a$ and $q = b$ in the given equation will give the required complete integral.

3.8b Solved-examples based on Art. 3.8a.

Ex. 1. Find a complete integral of $z = px + qy + p^2 + q^2$.

[Bilaspur 2000I; Kanpur 2009; Indore 2000; Jabalpur 2000;
K.U. Kurukshetra 2005; Ravishankar 2000; 04; Meerut 2010; Garhwal 2010]

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Sol. Let

$$f(x, y, z, p, q) = z - px - qy - p^2 - q^2 = 0 \quad \dots (1)$$

Charpit's auxiliary equations are $\frac{dp}{f_x + p f_z} = \frac{dq}{f_y + q f_z} = \frac{dz}{-p f_p - q f_q} = \frac{dx}{-f_p} = \frac{dy}{-f_q} \quad \dots (2)$

From (1), $f_x = -p$, $f_y = -q$, $f_z = 1$, $f_p = -x - 2p$ and $f_q = -y - 2q \quad \dots (3)$

Using (3), (2) reduces to

$$\frac{dp}{0} = \frac{dq}{0} = \frac{dz}{p(x+2p) + q(y+2q)} = \frac{dx}{x+2p} = \frac{dy}{y+2q} \quad \dots (4)$$

Taking the first fraction of (4), $dp = 0$ so that $p = a \quad \dots (5)$

Taking the second fraction of (4), $dq = 0$ so that $q = b \quad \dots (6)$

Putting $p = a$ and $q = b$ in (1), the required complete integral is

$$z = ax + by + a^2 + b^2, \quad a, b \text{ being arbitrary constants.}$$

Ex. 2. Find a complete integral of $q = 3p^2$. [Agra 2006]

Sol. Here given equation is

$$f(x, y, z, p, q) = 3p^2 - q = 0. \quad \dots (1)$$

$\therefore$ Charpit's auxiliary equations are $\frac{dp}{\frac{\partial f}{\partial x} + p \frac{\partial f}{\partial z}} = \frac{dq}{\frac{\partial f}{\partial y} + q \frac{\partial f}{\partial z}} = \frac{dz}{-p \frac{\partial f}{\partial p} - q \frac{\partial f}{\partial q}} = \frac{dx}{-\frac{\partial f}{\partial p}} = \frac{dy}{-\frac{\partial f}{\partial q}}$

or $\frac{dp}{0 + p \cdot 0} = \frac{dq}{0 + q \cdot 0} = \frac{dz}{-6p^2 + q} = \frac{dx}{-6p} = \frac{dy}{1}$, using (1) $\dots (2)$

Taking the first fraction of (1), $dp = 0$ so that $p = a. \quad \dots (3)$

Substituting this value of p in (1), we get $q = 3a^2. \quad \dots (4)$

Putting these values of p and q in $dz = p dx + q dy$, we get

$$dz = a dx + 3a^2 dy \quad \text{so that} \quad z = ax + 3a^2 y + b,$$

which is a complete integral, a and b being arbitrary constants.

Ex. 3. Find the complete integral of $zpq = p + q$ [Nagpur 2010; Meerut 2006]

Sol. Try yourself. Ans. $z^2 = 2(1+a)[x + (1/a)y] + b$, a, b being arbitrary constants

Ex. 4. Find a complete integral of $p^2 - y^2 q = y^2 - x^2$. [Delhi B.A. (Prog.) II, 2015]

Sol. Here given equation is $f(x, y, z, p, q) = p^2 - y^2 q - y^2 + x^2 = 0. \quad \dots (1)$

Charpit's auxiliary equations are $\frac{dp}{\frac{\partial f}{\partial x} + p \frac{\partial f}{\partial z}} = \frac{dq}{\frac{\partial f}{\partial y} + q \frac{\partial f}{\partial z}} = \frac{dz}{-p \frac{\partial f}{\partial p} - q \frac{\partial f}{\partial q}} = \frac{dx}{-\frac{\partial f}{\partial p}} = \frac{dy}{-\frac{\partial f}{\partial q}}$

or $\frac{dp}{2x} = \frac{dq}{-2qy - 2y} = \frac{dz}{-p(2p) - q(-q^2)} = \frac{dx}{-2p} = \frac{dy}{y^2}$, using (1) $\dots (2)$

Taking the first and fourth fractions, $p dp + x dx = 0$ so that $p^2 + x^2 = a^2 \quad \dots (3)$

Solving (1) and (3) for p and q , $p = (a^2 - x^2)^{1/2}$, $q = a^2 y^2 - 1.$

$\therefore dz = p dx + q dy = (a^2 - x^2)^{1/2} dx + (a^2 y^2 - 1) dy.$

Integrating, $z = (x/2) \times (a^2 - x^2)^{1/2} + (a^2/2) \times \sin^{-1}(x/a) - (a^2/y) - y + b.$

Ex. 5. Find a complete integral of $z^2(p^2 z^2 + q^2) = 1$. [Lucknow 2008; Kanpur 2015]

Sol. Here given equation is $f(x, y, z, p, q) = p^2 z^4 + q^2 z^2 - 1 = 0. \quad \dots (1)$

Charpit's auxiliary equations are $\frac{dp}{f_x + p f_z} = \frac{dq}{f_y + q f_z} = \frac{dz}{-p f_p - q f_q} = \frac{dx}{-f_p} = \frac{dy}{-f_q}$

or $\frac{dp}{p(4p^2 z^3 + 2z q^2)} = \frac{dq}{q(4p^2 z^3 + 2z q^2)} = \frac{dz}{-2p^2 z^4 - 2q^2 z^2} = \frac{dx}{-2p z^4} = \frac{dy}{-2q z^2}$, by (1) $\dots (2)$

Taking the first two fractions, $(1/p)dp = (1/q)dq$ so that $p = aq.$

Solving (1) and (2) for p and q ,
$$p = \frac{a}{z(a^2z^2+1)^{1/2}}, \quad q = \frac{1}{z(a^2z^2+1)^{1/2}}.$$

$\therefore dz = p dx + q dy = (a dx + dy)/z(a^2z^2+1)^{1/2}$ or $adx + dy = z(a^2z^2+1)^{1/2} dz.$

Integrating, $ax + y = \int (a^2z^2+1)^{1/2} \cdot z dz \dots (3)$

Putting $a^2z^2+1 = t^2$ so that $2a^2z dz = 2t dt$, (3) becomes

$ax + y = \int (1/a^2) t \cdot t dt$ or $ax + y + b = (1/3a^2)t^3$, where $t = (a^2z^2+1)^{1/2}$

or $ax + y + b = (1/3a^2) \times (a^2z^2+1)^{3/2}$ or $9a^4(ax+y+b)^2 = (a^2z^2+1)^3$, which is a complete integral, a and b being arbitrary constants.

Ex. 6. Find a complete integral of $px + qy = pq$. [Kurukshetra 2006 Rajasthan 2000, 01, Avadh 2010; Gulbarga 2005; Meerut 2013, 17; Kanpur 2004; Jiwaji 2004; Kanpur 2012; Rohilkhand 2010; Agra 2012; Bhopal 2010]

Sol. Here given equation is $f(x, y, z, p, q) \equiv px + qy - pq = 0$ (1)

Charpit's auxiliary equations are
$$\frac{dp}{f_x + p f_z} = \frac{dq}{f_y + q f_z} = \frac{dz}{-p f_p - q f_q} = \frac{dx}{-f_p} = \frac{dy}{-f_q}$$

or
$$\frac{dp}{-(x-q)} = \frac{dy}{-(y-q)} = \frac{dz}{-p(x-q) - q(y-p)} = \frac{dp}{p+p \cdot 0} = \frac{dq}{q+q \cdot 0}, \text{ by (1)} \dots (2)$$

Taking the last two fractions of (2),

$(1/p)dp = (1/q)dq.$

Integrating, $\log p = \log q + \log a$ or $p = aq$ (3)

Substituting this value of p in (1), we have

$aqx + qy - aq^2 = 0$ or $aq = ax + y$, as $q \neq 0$... (4)

$\therefore$ From (3) and (4), $q = (ax + y)/a$ and $p = ax + y$ (5)

Putting these values of p and q in $dz = p dx + q dy$, we get

$dz = (ax + y)dx + [(ax + y)/a] dy$ or $adz = (ax + y)(adx + dy)$

or $adz = (ax + y) d(ax + y) = u du$, where $u = ax + y$.

Integrating,

$az = u^2/2 + b = (ax + y)^2/2 + b$,

which is a complete integral, a and b being arbitrary constants.

Ex. 7. Find the complete integrals of following equations:

(i) $q = (z + px)^2$

[Indore 2004; Ravishanker 2005]

(ii) $p = (z + qy)^2$

[Meerut 2015; Agra 2001; Delhi B.Sc. (Prog) 2008;

Delhi B.A. (Prog.) II, 2014, 16, 18; Lucknow 2011; Kurukshetra 2005]

Sol. (i). Here given equations is $f(x, y, z, p, q) = (z + px)^2 - q = 0$... (1)

Charpit's auxiliary equations are
$$\frac{dp}{f_x + p f_z} = \frac{dq}{f_y + q f_z} = \frac{dz}{-p f_p - q f_q} = \frac{dx}{-f_p} = \frac{dy}{-f_q}$$

or
$$\frac{dp}{2p(z+px)+2p(z+px)} = \frac{dq}{2q(z+px)} = \frac{dz}{-2px(z+px)+q} = \frac{dx}{-2x(z+px)} = \frac{dy}{0}, \text{ by (1)}$$

Taking the second and fourth fractions,

$(1/q)dq = -(1/x)dx.$

Integrating, $\log q = \log a - \log x$ so that

$q = a/x$ (2)

Substituting the above value of q in (1), we have

$(z + px)^2 = a/x$ or $px = \sqrt{a}/\sqrt{x} - z$ or $p = \sqrt{a}/x\sqrt{x} - z/x$ (3)

$\therefore dz = p dx + q dy = \left(\frac{\sqrt{a}}{x\sqrt{x}} - \frac{z}{x} \right) dx + \frac{a}{x} dy$, by (2) and (3)

or $x dz = \sqrt{a} x^{-1/2} dx - z dx + a dy$ or $x dz + z dx = \sqrt{a} x^{-1/2} dx + a dy$

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or
$$d(xz) = \sqrt{a} x^{1/2} dx + ady.$$

Integrating, $xz = 2\sqrt{a}\sqrt{x} + ay + b$, a, b being arbitrary constants

(ii) Sol. Do as in part (1). Ans. $yz = ax + \sqrt{ay} + b$.

Ex. 8(a). Find a complete integral of $(p^2 + q^2)x = pz$. [CSIR-UGC (NET), June 2017]

[Agra 2003; Rajasthan 2005; Ravishankar 2001; Delhi B.A. (Prog.) II, 2013]

(b). Find the complete integral of the partial differential equation $(p^2 + q^2)x = pz$ and deduce the solution which passes through the curve $x = 0, z^2 = 4y$. [Meerut 2007]

Sol. Let
$$f(x, y, q, p, q) = (p^2 + q^2)x - pz = 0. \quad \dots(1)$$

Charpit's auxiliary equations are
$$\frac{dp}{f_x + p f_z} = \frac{dq}{f_y + q f_z} = \frac{dz}{-p f_p - q f_q} = \frac{dx}{-f_p} = \frac{dy}{-f_q}$$

giving $dp/q^2 = dq/(-pq)$, by (1) or $2pdp + 2q dq = 0$.

Integrating, $p^2 + q^2 = a^2$, where a is an arbitrary constant. ... (2)

Solving (1) and (2), $p = a^2 x/q$ and $q = (a/z) \times \sqrt{(z^2 - a^2 x^2)}$ (3)

$\therefore dz = p dx + q dy = \frac{a^2 x dx}{z} + \frac{a \sqrt{(z^2 - a^2 x^2)} dy}{z}$ or $\frac{z dz - a^2 x dx}{\sqrt{(z^2 - a^2 x^2)}} = a dy$.

Putting $z^2 - a^2 x^2 = t$ so that $2(z dz - a^2 x dx) = dt$, we get

$(1/2\sqrt{t}) dt = a dy$ or $(1/2) \times t^{-1/2} = a dy$.

Integrating, $t^{1/2} = ay + b$ or $\sqrt{(z^2 - a^2 x^2)} = ay + b$, as $t = \sqrt{z^2 - a^2 x^2}$

or $z^2 - a^2 x^2 = (ay + b)^2$ or $z^2 = a^2 x^2 + (ay + b)^2$ (4)

(b) Proceeding as in part (a), (4) is the complete integral.

The parametric equations of the given curve $x = 0, z^2 = 4y$ are given by

$x = 0, y = t^2, z = 2t$... (5)

Therefore the intersections of (1) and (2) are determined by

$4t^2 = (a^2 t^2 + b)^2$ or $a^2 t^4 + 2(ab - 2)t^2 + b^2 = 0$... (6)

Equation (6) has equal roots if its discriminant = 0, i.e., if

$4(ab - 2)^2 - 4a^2 b^2 = 0$ or $a^2 b^2 = 1$ so that $b = 1/a$

Hence from (4), the appropriate one parameter sub-system is given by

$z^2 = a^2 x^2 + (ay + 1/a)^2$ or $a^4(x^2 + y^2) + a^2(2y - z^2) + 1 = 0$,

which is a quadratic equation in parameter ' a '. Therefore, this has for its envelope surface

$(2y - z^2)^2 - 4(x^2 + y^2) = 0$ or $(2y - z^2)^2 = 4(x^2 + y^2)$... (7)

The desired solution is given by the function z defined by equation (7).

Ex. 8(c). Find a complete, singular and general integrals of $(p^2 + q^2)y = qz$.

[Guwahati 2007; Agra 2001; Lucknow 2008; Delhi B.Sc. (Prog.) II 2012; Garhwal 2005;

Meerut 2014, 16; K.V. Kurukshetra 2004; Kanpur 2014; Rohilkhand 2001; Pune 2010]

Sol. Here the given equation is
$$f(x, y, z, p, q) = (p^2 + q^2)y - qz = 0. \quad \dots(1)$$

Charpit's auxiliary equations are
$$\frac{dp}{f_x + p f_z} = \frac{dq}{f_y + q f_z} = \frac{dz}{-p f_p - q f_q} = \frac{dx}{-f_p} = \frac{dy}{-f_q}$$

or
$$\frac{dp}{-pq} = \frac{dq}{p^2} = \frac{dz}{-2p^2 y + qz - 2q^2 y} = \frac{dx}{-2py} = \frac{dy}{-2qy + z}, \text{ by (1)} \quad \dots(2)$$

Taking the first two fractions, we get $2pdp + 2q dq = 0$ so that $p^2 + q^2 = a$... (3)

Using (3), (1) gives $a^2 y = qz$ or $q = a^2 y/z$.

Putting this value of q in (3), we get

$$p = \sqrt{(a^2 - q^2)} = \sqrt{a^2 - (a^4 y^2 / z^2)} = \frac{a}{z} \sqrt{(z^2 - a^2 y^2)}.$$

Now putting these values of p and q in $dz = p dx + q dy$, we have

$$dz = \frac{a}{z} \sqrt{(z^2 - a^2 y^2)} dx + \frac{a^2 y dy}{z} \quad \text{or} \quad \frac{z dz - a^2 y dy}{\sqrt{(z^2 - a^2 y^2)}} = a dx.$$

Integrating, $(z^2 - a^2 y^2)^{1/2} = ax + b$ or $z^2 - a^2 y^2 = (ax + b)^2$, ... (4)
which is a required complete integral, a, b being arbitrary constants.

Singular Integral. Differentiating (4) partially w.r.t. a and b , we have

$$0 = 2ay^2 + 2(ax + b)x \quad \dots (5)$$

$$\text{and} \quad 0 = 2(ax + b). \quad \dots (6)$$

Eliminating a and b between (4), (5) and (6), we get $z = 0$ which clearly satisfies (1) and hence it is the singular integral.

General Integral. Replacing b by $\phi(a)$ in (4), we get

$$z^2 - a^2 y^2 = [ax + \phi(a)]^2. \quad \dots (7)$$

$$\text{Differentiating (7) partially w.r.t. } a, \quad -2ay^2 = 2[ax + \phi(a)] \cdot [x + \phi'(a)]. \quad \dots (8)$$

General integral is obtained by eliminating a from (7) and (8).

Ex. 9. Find a complete integral of $p(1 + q^2) + (b - z)q = 0$.

Sol. Try yourself. **Ans.** $4\{c(z - b) - 1\} = (x + cy + a)^2$
which is a complete integral, a and c being arbitrary constants.

Ex. 10. Find a complete and singular integrals of $2xz - px^2 - 2qxy + pq = 0$.

[Delhi B.Sc. (Hons.) 2001, 01, 05; Kanpur 2001, 03; Meerut 2017; Bhopal 2004, 10;

M.D.U. Rohtak 2004, Ravishanker 2004; Rajasthan 2000, 03, 05, 10]

Sol. Here given equation is $f(x, y, z, p, q) = 2xz - px^2 - 2qxy + pq = 0$ (1)

$$\text{Charpit's auxiliary equations are} \quad \frac{dp}{\frac{\partial f}{\partial x} + p \frac{\partial f}{\partial z}} = \frac{dq}{\frac{\partial f}{\partial y} + q \frac{\partial f}{\partial z}} = \frac{dz}{-p \frac{\partial f}{\partial p} - q \frac{\partial f}{\partial q}} = \frac{dx}{-\frac{\partial f}{\partial p}} = \frac{dy}{-\frac{\partial f}{\partial q}}$$

$$\text{or} \quad \frac{dp}{2z - 2qy} = \frac{dq}{0} = \frac{dx}{x^2 - q} = \frac{dy}{2xy - p} = \frac{dz}{px^2 + 2xyq - 2pq}, \text{ by (1)}$$

The second fraction gives $dq = 0$ so that $q = a$

Putting $q = a$ in (1), we get $p = 2x(z - ay)/(x^2 - a)$

Putting values p and q in $dz = p dx + q dy$, we get

$$dz = \frac{2x(z - ay)}{x^2 - a} dx + a dy \quad \text{or} \quad \frac{dz - a dy}{z - ay} = \frac{2x dx}{x^2 - a}.$$

$$\text{Integrating,} \quad \log(z - ay) = \log(x^2 - a) + \log b \quad \text{or} \quad z - ay = b(x^2 - a), \quad \dots (2)$$

which is the complete integral, a and b being arbitrary constants.

Differentiating (2) partially with respect to a and b , we get

$$0 = y - b \quad \text{and} \quad 0 = x^2 - a. \quad \dots (3)$$

$$\text{Solving (3) for } a \text{ and } b, \quad a = x^2 \quad \text{and} \quad b = y. \quad \dots (4)$$

Substituting the values of a and b given by (4) in (2), we get $z = x^2 y$, which is the required singular integral.

Ex. 11. Find a complete integrals of the following partial differential equations:

(i) $q = px + p^2$. [Delhi B.A. (Prog.) II, 2014; Sagar 2003]

(ii) $q = -px + p^2$.

Sol. (i) Here given equation is $f(x, y, z, p, q) \equiv q - px - p^2 = 0$ (1)

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Charpit's auxiliary equations are $\frac{dp}{f_x + p f_z} = \frac{dq}{f_y + q f_z} = \frac{dz}{-p f_p - q f_q} = \frac{dx}{-f_p} = \frac{dy}{-f_q}$

or $\frac{dp}{-p} = \frac{dq}{0} = \frac{dz}{-p(-x-2p)-q} = \frac{dx}{-(-x-2p)} = \frac{dy}{-1}$, by (1)

The 2nd fraction gives $dq = 0$ so that $q = a$.

Putting $q = a$ in (1) gives $p^2 + px - a = 0$ so that $p = (1/2) \times [-x \pm (x^2 + 4a)^{1/2}]$

Putting these values of p and q in $dz = p dx + q dy$, we get

$$dz = -(x/2) \times dx \pm (1/2) \times (x^2 + 4a)^{1/2} dx + a dy.$$

Integrating, the required complete integral is

$$z = -\frac{x^2}{4} \pm \frac{1}{2} \left[\frac{x}{2} (x^2 + 4a)^{1/2} + 2a \log \{x + (x^2 + 4a)^{1/2}\} \right] + ay + b,$$

Part (ii). Proceed like part (i) yourself. Complete integral is

$$z = \frac{x^2}{4} \pm \frac{1}{2} \left[\frac{x}{2} (x^2 + 4a)^{1/2} + 2a \log \{x + (x^2 + 4a)^{1/2}\} \right] + ay + b.$$

Ex. 12. Find a complete integral of $pxy + pq + qy = yz$. [Delhi B.A. (Prog) H 2010, 15]
Agra 2011; Avadh 2009; Meerut 2015, 16; Kanpur 2013]

Sol. Given $f(x, y, z, p, q) \equiv pxy + pq + qy - yz = 0$ (1)

Charpit's auxiliary equation are $\frac{dp}{f_x + p f_z} = \frac{dq}{f_y + q f_z} = \frac{dz}{-p f_p - q f_q} = \frac{dx}{-f_p} = \frac{dy}{-f_q}$

or $\frac{dp}{0} = \frac{dq}{(px+q)+qy} = \frac{dz}{-p(xy+q)-q(p+y)} = \frac{dx}{-(xy+q)} = \frac{dy}{-(p+y)}$, by (1)

The first fraction gives $dp = 0$ so that $p = a$.

Putting $p = a$ in (1) gives $axy + aq + qy = yz$ so that $q = y(z - ax)/(a + y)$.

Putting these values of p and q in $dz = p dx + q dy$, we get

$$dz = a dx + \frac{y(z - ax)}{a + y} dy \quad \text{or} \quad \frac{dz - a dx}{z - ax} = \frac{y dy}{a + y} = \left(1 - \frac{a}{a + y}\right) dy.$$

Integrating, $\log(z - ax) = y - a \log(a + y) + \log b$, a, b , being arbitrary constants.

or $\log(z - ax) + \log(a + y)^a - \log b = y$ or $(z - ax)(a + y)^a = be^y$

Ex. 13. Find a complete integral $p^2 + q^2 - 2px - 2qy + 1 = 0$.

[Patna 2003; Meerut 2003; Ravishanker 2010]

Sol. Given $f(x, y, z, p, q) \equiv p^2 + q^2 - 2px - 2qy + 1 = 0$ (1)

Charpit's auxiliary equations are $\frac{dp}{f_x + p f_z} = \frac{dq}{f_y + q f_z} = \frac{dz}{-p f_p - q f_q} = \frac{dx}{-f_p} = \frac{dy}{-f_q}$

or $\frac{dp}{-2p} = \frac{dq}{-2q} = \frac{dz}{-p(2p-2x)-q(2q-2y)} = \frac{dx}{-(2p-2x)} = \frac{dy}{-(2q-2y)}$, by (1)

The first two fractions give $(1/p)dp = (1/q)dq$ so that $p = aq$.

Putting $p = aq$ in (1), $a^2 q^2 + q^2 - 2aqx - 2qy + 1 = 0$ or $(a^2 + 1)q^2 - 2(ax + y)q + 1 = 0$.

$$\Rightarrow q = \frac{2(ax + y) \pm \sqrt{4(ax + y)^2 - 4(a^2 + 1)}}{2(a^2 + 1)}, \quad p = aq = a \frac{2(ax + y) \pm \sqrt{4(ax + y)^2 - 4(a^2 + 1)}}{2(a^2 + 1)}.$$

Putting these values of p and q in $dz = p dx + q dy$, we get

$$dz = \frac{(ax+y) \pm \sqrt{\{(ax+y)^2 - (a^2+1)\}}}{(a^2+1)} (adx + dy). \quad \dots(2)$$

Put $ax + y = v$ so that $a dx + dy = dv$. Then (2) gives

$$(a^2 + 1) dz = [v \pm \sqrt{v^2 - (a^2 + 1)}] dv.$$

Integrating, $(a^2 + 1) z = v^2/2 \pm [(v/2) \times \sqrt{v^2 - (a^2 + 1)} - (1/2) \times (a^2 + 1) \log (v + \sqrt{v^2 - (a^2 + 1)})] + b$

is the complete integral, where $v = ax + b$ and a, b are arbitrary constants.

Ex. 14. Find a complete integral of $p^2 + q^2 - 2px - 2qy + 2xy = 0$. [Meerut 2014]

Sol. Given equation is $f(x, y, z, p, q) \equiv p^2 + q^2 - 2px - 2qy + 2xy = 0$. $\dots(1)$

Charpit's auxiliary equations are $\frac{dp}{\frac{\partial f}{\partial x} + p \frac{\partial f}{\partial z}} = \frac{dq}{\frac{\partial f}{\partial y} + q \frac{\partial f}{\partial z}} = \frac{dz}{-p \frac{\partial f}{\partial p} - q \frac{\partial f}{\partial q}} = \frac{dx}{-\frac{\partial f}{\partial p}} = \frac{dy}{-\frac{\partial f}{\partial q}}$

or $\frac{dp}{-2p+2y} = \frac{dq}{-2q+2x} = \frac{dx}{2x-2p} = \frac{dy}{2y-2q}$, by (1)

which gives $\frac{dp+dq}{2(x+y-p-q)} = \frac{dx+dy}{2(x+y-p-q)}$

or $dp + dq = dx + dy$ i.e., $dp - dx + dq - dy = 0$.

Integrating, $(p-x) + (q-y) = a$ $\dots(2)$

Re-writing (1), $(p-x)^2 + (q-y)^2 = (x-y)^2$. $\dots(3)$

Putting the value of $(q-y)$ from (2) in (3), we get

$(p-x)^2 + [a - (p-x)]^2 = (x-y)^2$ or $2(p-x)^2 - 2a(p-x) + \{a^2 - (x-y)^2\} = 0$.

$\therefore p-x = \frac{2a \pm \sqrt{4a^2 - 4.2.\{a^2 - (x-y)^2\}}}{4} \Rightarrow p = x + \frac{1}{2} [a \pm \sqrt{2(x-y)^2 - a^2}]$,

$\therefore$ (2) gives $q = a + y - p + x$ or $q = y + (1/2) \times [a \mp \sqrt{2(x-y)^2 - a^2}]$.

Putting these value of p and q in $dz = p dx + q dy$, we get

$$dz = x dx + y dy + (a/2) \times (dx + dy) \pm (1/2) \sqrt{2(x-y)^2 - a^2} (dx - dy)$$

or $dz = x dx + y dy + \frac{a}{2} (dx + dy) \pm \frac{1}{\sqrt{2}} \sqrt{(x-y)^2 - a^2/2} (dx - dy)$.

Integrating, the desired complete integral is

$$z = \frac{x^2 + y^2}{2} + \frac{a(x+y)}{2} \pm \frac{1}{\sqrt{2}} \left(\frac{x-y}{2} \sqrt{(x-y)^2 - a^2/2} - \frac{a^2}{4} \log \left[(x-y) + \sqrt{(x-y)^2 - a^2/2} \right] \right)$$

Ex. 15. Find a complete integral of $p^2x + q^2y = z$. [Gujarat 2005; K.U. Kurukshetra 2001; Meerut 2008; Agra 2004; Lucknow 2009; Delhi B.A. (Prog.) II 2012; Punjab 2001]

Sol. Given equation is $f(x, y, z, p, q) \equiv p^2x + q^2y - z = 0$. $\dots(1)$

Charpit's auxiliary equations are $\frac{dp}{f_x + p f_z} = \frac{dq}{f_y + q f_z} = \frac{dz}{-p f_p - q f_q} = \frac{dx}{-f_p} = \frac{dy}{-f_q}$

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or
$$\frac{dp}{-p+p^2} = \frac{dq}{-q+q^2} = \frac{dz}{-2(p^2x+q^2y)} = \frac{dx}{-2px} = \frac{dy}{-2qy}, \text{ by (1)} \quad \dots(2)$$

Now, each fraction in (2)
$$= \frac{2px dp + p^2 dx}{2px(-p+p^2) + p^2(-2px)} = \frac{2qy dq + q^2 dy}{2qy(-q+q^2) + q^2(-2qy)}$$

or
$$\frac{d(p^2x)}{-2p^2x} = \frac{d(q^2y)}{-2qy} \quad \text{i.e.,} \quad \frac{d(p^2x)}{p^2x} = \frac{d(q^2y)}{q^2y}.$$

Integrating it, $\log(p^2x) = \log(q^2y) + \log a$ or $p^2x = q^2ya. \quad \dots(3)$

Form (1) and (3), $aq^2y + q^2y = z$ or $q = [z/(1+a)]^{1/2}. \quad \dots(4)$

Form (3) and (4),
$$p = q \left(\frac{ya}{x} \right)^{1/2} = \left\{ \frac{za}{(1+a)x} \right\}^{1/2}.$$

Putting the above values of p and q in $dz = p dx + q dy$, we get

$$dz = \left\{ \frac{za}{(1+a)x} \right\}^{1/2} dx + \left\{ \frac{z}{(1+a)y} \right\}^{1/2} dy \quad \text{or} \quad (1+a)^{1/2} z^{1/2} dz = \sqrt{ax}^{-1/2} dx + y^{-1/2} dy.$$

Integrating, $(1+a)^{1/2} \sqrt{z} = \sqrt{a}\sqrt{x} + \sqrt{y} + b$, a, b being arbitrary constants.

Ex. 16. Find a complete integral of $2z + p^2 + qy + 2y^2 = 0$. [Meerut 2000;

Bilaspur 2004, M.D.U Rohtak 2005; Ranchi 2010]

Sol. Given equation is $f(x, y, z, p, q) = 2z + p^2 + qy + 2y^2 = 0. \quad \dots(1)$

Charpit's auxiliary equations are
$$\frac{dp}{f_x + p f_z} = \frac{dq}{f_y + q f_z} = \frac{dz}{-p f_p - q f_q} = \frac{dp}{-f_p} = \frac{dq}{-f_q}$$

or
$$\frac{dp}{0+2p} = \frac{dq}{(q+4y)+2q} = \frac{dz}{-p \times (2p) - qy} = \frac{dx}{-2p} = \frac{dy}{-y}, \text{ by (1)}$$

Taking the first and fourth fractions, $dp = -dx.$

Integrating, $p = a - x$ or $p = -(x - a). \quad \dots(2)$

Using (2), (1) becomes $2z + (a - x)^2 + qy + 2y^2 = 0$

$\therefore q = -[2z + (x - a)^2 + 2y^2]/y. \quad \dots(3)$

$\therefore dz = p dx + q dy = -(x - a) dx - [2z + (x - a)^2 + 2y^2]/y dy, \text{ by (2) and (3)}$

Multiplying both sides by $2y^2$ and re-writing, we have

or
$$2y^2 dz = -2(x - a)y^2 dx - 4zy dy - 2y(x - a)^2 dy - 4y^3 dy$$

or
$$2(y^2 dz + 2zy dy) + [2(x - a)^2 y^2 dx + 2y(x - a)^2 dy] + 4y^3 dy = 0$$

Integrating, $2y^2 z + y^2(x - a)^2 + y^4 = b$, a, b being arbitrary constants

Ex. 17(a). Find a complete integral of $z^2 = pqxy$ [Lucknow 2010]

Sol. Try yourself. **Ans.** $z = b x^a y^{1/a}$, a and b being arbitrary constants

Ex. 17(b). Find a complete integral of $2(z + px + qy) = yp^2$. [Lucknow 2010;

Delhi B.A. (Prog.) II 2007, 10; CDLU 2004; Delhi B.Sc. (Hons.) 2008]

Sol. Given equation is $f(x, y, z, p, q) = 2(z + px + qy) - yp^2 = 0 \quad \dots(1)$

Charpit's auxiliary equations are
$$\frac{dp}{f_x + p f_z} = \frac{dq}{f_y + q f_z} = \frac{dz}{-p f_p - q f_q} = \frac{dp}{-f_p} = \frac{dq}{-f_q}$$

or
$$\frac{dp}{2p+2p} = \frac{dq}{2q-p^2+2q} = \frac{dz}{-p(2x-2yp)-q \times 2y} = \frac{dx}{-(2x-2yp)} = \frac{dy}{-2y}, \text{ by (1)}$$

Taking the first and the last fractions,
$$\frac{dp}{4p} = \frac{dy}{-2y} \quad \text{or} \quad \frac{dp}{p} + 2 \frac{dy}{y} = 0.$$

Integrating, $\log p + 2 \log y = \log a$ or $py^2 = a$ (2)

Solving (1) and (2) for p and q , $p = \frac{a}{y^2}$ and $q = -\frac{z}{y} - \frac{ax}{y^3} + \frac{a^2}{2y^4}$.

$$\therefore dz = p dx + q dy = \frac{a}{y^2} dx + \left[-\frac{z}{y} - \frac{ax}{y^3} + \frac{a^2}{2y^4} \right] dy$$

Multiplying both sides by y and re-arranging, we get

$$(ydz + zdy) - a \left(\frac{y dx - x dy}{y^2} \right) - \frac{a^2}{2y^3} dy = 0 \quad \text{or} \quad d(yz) - ad \left(\frac{x}{y} \right) - \frac{a^2}{2} y^{-3} dy = 0.$$

Integrating, $yz - a(x/y) + (a^2/4y^2) = b$, a, b being arbitrary constants. ... (3)

Ex. 18. Using Charpit's method, find three complete integrals of $pq = px + qy$.

(Kanpur 2004; Meerut 2002; Rajasthan 2001)

Sol. Here given equation is $f(x, y, z, p, q) = pq - px - qy = 0$ (1)

Charpit's auxiliary equations are $\frac{dp}{f_x + p f_z} = \frac{dq}{f_y + q f_z} = \frac{dz}{-p f_p - q f_q} = \frac{dx}{-f_p} = \frac{dy}{-f_q}$

or $\frac{dp}{-p} = \frac{dq}{-q} = \frac{dz}{-p(q-x) - p(p-y)} = \frac{dx}{-(q-x)} = \frac{dy}{-(p-y)}$, by (1) ... (2)

To find first complete integral. Taking the first two fractions of (2), we get

$$(1/p)dp = (1/q)dq \quad \text{so that} \quad \log p = \log q + \log a \quad \text{or} \quad p = aq. \quad \dots (3)$$

$$\text{Using (3),} \quad (1) \Rightarrow aq^2 = q(ax + y) \Rightarrow q = (ax + y)/a. \quad \dots (4)$$

$$\text{Hence, from (3), we have} \quad p = ax + y. \quad \dots (5)$$

$$\therefore dz = p dx + q dy = (ax + y)dx + [(ax + y)/a]dy = (1/a)(ax + y)(a dx + y).$$

Putting $ax + y = t$ so that $adx + dy = dt$, we get

$$dz = (1/a) \times t dt \quad \text{so that} \quad z = (1/2a) \times t^2 + b \quad \text{or} \quad z = (1/2a) \times (ax + y)^2 + b, \text{ as } t = ax + y.$$

To find second complete integral. Taking the second and the fourth ratios in (2), we get

$$dx/(q-x) = dq/q \quad \text{or} \quad q dx + x dq = q dq.$$

$$\text{Integrating,} \quad qx = q^2/2 + a/2 \quad \text{or} \quad q^2 - 2xq + a = 0.$$

$$\therefore q = [2x \pm 2(x^2 - a)^{1/2}]/2 \quad \text{so that} \quad q = x + (x^2 - a)^{1/2}. \quad \dots (6)$$

$$\text{Using (6),} \quad (1) \Rightarrow p[x + (x^2 - a)^{1/2}] - px - y[x + (x^2 - a)^{1/2}] = 0$$

so that $p = \{1 + x/(x^2 - a)^{1/2}\}y$ (7)

$$\therefore dz = p dx + q dy = \left\{ 1 + x/(x^2 - a)^{1/2} \right\} y dx + [x + (x^2 - a)^{1/2}] dy$$

or $dz = (y dx + x dy) + \left[\frac{xy dy}{(x^2 - a)^{1/2}} + (x^2 - a)^{1/2} dy \right] \quad \text{or} \quad dz = d(xy) + d[y(x^2 - a)^{1/2}].$

Integrating, $z = xy + y(x^2 - a)^{1/2} + b$, a, b being arbitrary constants.

To find third complete integral. Taking the first and the fifth ratios of (2) and proceeding as above third complete integral is $z = xy + x(y^2 - a)^{1/2} + b$.

Ex. 19. Find complete integral of $xp + 3yq = 2(z - x^2q^2)$. [Delhi B.Sc. (Prog) II 2009; Meerut 2013; Delhi B.Sc. (Hons) II 2010;]

Sol. Given equation is $f(x, y, z, p, q) = xp + 3yq - 2z + 2x^2q^2 = 0$ (1)

Charpit's auxiliary equations are $\frac{dp}{f_x + p f_z} = \frac{dq}{f_y + q f_z} = \frac{dz}{-p f_p - q f_q} = \frac{dx}{-f_p} = \frac{dy}{-f_q}$

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or
$$\frac{dp}{-p+4xq^2} = \frac{dq}{q} = \frac{dz}{-px-q(3y+4x^2q)} = \frac{dx}{-x} = \frac{dy}{-3y-4x^2q}, \text{ by (1)} \quad \dots(2)$$

(2) $\Rightarrow \frac{dq}{q} = \frac{dx}{-x} \Rightarrow \log q = \log a - \log x \Rightarrow qx = a \Rightarrow q = \frac{a}{x} \quad \dots(3)$

Using (3), (1) $\Rightarrow xp + 3y(a/x) - 2z + 2x^2(a^2/x^2) = 0 \Rightarrow p = \frac{2(z-a^2)}{x} - \frac{3ay}{x^2} \quad \dots(4)$

$\therefore dz = p dx + q dy = \left\{ \frac{2(z-a^2)}{x} - \frac{3ay}{x^2} \right\} dx + \frac{a}{x} dy$

or $x^2 dz = 2x(z-a^2)dx - 3ay dx + ax dy$ or $x^2 dz - 2x(z-a^2)dx = -3ay dx + ax dy$

or $\frac{x^2 dz - 2x(z-a^2)dx}{x^4} = -\frac{3ay dx}{x^4} + \frac{a dy}{x^3}$ or $d\left(\frac{z-a^2}{x^2}\right) = d\left(\frac{ay}{x^3}\right)$

Integrating, $(z-a^2)/x^2 = (ay)/x^3 + b$ or $z = a(a+y/x) + bx^2$.

Ex. 20 Find a complete integral of $z = pq$. [Sagar 2004, Ravishankar 2003, Rewa 2003; Rohilkhand 2009]

Sol. Here given equation is $f(x, y, z, p, q) = z - pq = 0 \quad \dots (1)$

Charpit's auxiliary equations are $\frac{dp}{f_x + p f_z} = \frac{dq}{f_y + q f_z} = \frac{dz}{-p f_p - q f_q} = \frac{dx}{-f_p} = \frac{dy}{-f_q}$

or $\frac{dp}{-p} = \frac{dq}{-q} = \frac{dz}{-2pq} = \frac{dx}{-q} = \frac{dy}{-p}, \text{ by (1)} \quad \dots (2)$

Taking the first and last fractions of (2), $dp = dy$

Integrating $p = y + a$, a being an arbitrary constant $\dots (3)$

Similarly, taking the second and fourth fractions of (2), we get

$dq = dx$ so that $q = x + b$, b being an arbitrary constant. $\dots (4)$

Putting values of p and q given by (3) and (4) in (1), we get

$z = (x+b)(x+a)$, which is the required complete integral.

Ex. 21. Find the complete integral of $(p+q)(px+qy)=1$.

[Kanpur 2010; Meerut 2007, 12; Delhi B.Sc. (Hons.) 2007, Purvanchal 2007]

Sol. Let $f(x, y, z, p, q) = (p+q)(px+qy)-1=0 \quad \dots (1)$

Charpit's auxiliary equations $\frac{dp}{f_x + p f_z} = \frac{dq}{f_y + q f_z} = \frac{dz}{-p f_p - q f_q} = \frac{dx}{-f_p} = \frac{dy}{-f_q}$

give $\frac{dp}{p(p+q)} = \frac{dq}{q(p+q)} = \dots$ so that $\frac{dp}{p} = \frac{dq}{q}$, using (1)

Integrating, $p = aq$, a being an arbitrary constant $\dots (2)$

Putting $p = aq$ in (2) gives $(aq+q)(aqx+qy)-1=0$ or $q^2(1+a)(ax+y)=1 \quad \dots (3)$

$\therefore$ From (2) and (3), $q = 1/(1+a)^{1/2}(ax+y)^{1/2}$, $p = a/(1+a)^{1/2}(ax+y)^{1/2}$

Putting these values of p and q in $dz = p dx + q dy$, we get

$$dz = \frac{a dx}{(1+a)^{1/2}(ax+y)^{1/2}} + \frac{dy}{(1+a)^{1/2}(ax+y)^{1/2}} = \frac{d(ax+y)}{(1+a)^{1/2}(ax+y)^{1/2}}$$

Integrating, $z(1+a)^{1/2} = 2(ax+y)^{1/2} + b$, a, b being arbitrary constants.

Ex. 22. Find the complete integral of the following partial differential equations

(a) $px^5 - 4q^2x^2 + 6x^2z - 2 = 0$. [Delhi 2002, 2011; Meerut 2012]

(b) $px^5 - 4q^3x^2 + 6x^2z - 2 = 0$

Sol. (a) Let $f(x, y, z, p, q) = px^5 - 4q^2x^2 + 6x^2z - 2 = 0$... (1)

Charpit's auxiliary equations are $\frac{dp}{f_x + p f_z} = \frac{dq}{f_y + q f_z} = \frac{dz}{-p f_p - q f_q} = \frac{dx}{-f_p} = \frac{dy}{-f_q}$

or $\frac{dp}{5px^4 - 8q^2x + 12xz + 6px^2} = \frac{dq}{-6qx^2} = \frac{dz}{-px^5 + 8q^2x^2} = \frac{dx}{-x^5} = \frac{dy}{8qx^2}$, by (1)

Taking the second and the last fractions, $4dq = 3dy$

Integrating, $4q = 3y + 3a$ or $q = 3(y+a)/4$... (2)

Using (2), (1) gives $p = \{(9/4) \times (y+a)^2 - 6x^2z + 2\} / x^5$... (3)

Putting the above values of p and q in $dz = p dx + q dy$, we get

$$dz = (9/4x^3)(y+a)^2 dx - (6z/x^3) dx + (2/x^5)dx + (3/4)(y+a)dy$$

or $(6z/x^3)dx + dz = \{(9/4x^3)(y+a)^2 dx + (3/4)(y+a) dy\} + (2/x^5)dx$... (4)

The total differential equation (4) is always integrable. To solve (4), we first proceed to find the integrating factor of the L.H.S. of (4). Comparing L.H.S. of (4) with $M dx + N dz$ (here on L.H.S. we have variable x, z in place of usual variables x, y), we have $M = 6z/x^3$ and $N = 1$.

$$\frac{1}{N} \left(\frac{\partial M}{\partial z} - \frac{\partial N}{\partial x} \right) = \frac{6}{x^3}, \text{ which is function } x \text{ alone and so I.F.} = e^{\int (6/x^3) dx} = e^{-3/x^2}.$$

Multiplying both sides of (4) by I.F. e^{-3/x^2} , we get

$$(6z/x^3) e^{-3/x^2} dx + e^{-3/x^2} dz = (3/8) \times \{(6/x^3)(y+a)^2 e^{-3/x^2} dx + 2(y+a) e^{-3/x^2} dy\} + (2/x^5) e^{-3/x^2} dx$$

or $d(z e^{-3/x^2}) = (3/8) \times d\{(y+a)^2 e^{-3/x^2}\} + (2/x^5) \times e^{-3/x^2} dx$

Integrating, $z e^{-3/x^2} = (3/8) \times (y+a)^2 e^{-3/x^2} + 2 \int (1/x^2) e^{-3/x^2} (1/x^3) dx$

or $z e^{-3/x^2} = (3/8) \times (y+a)^2 e^{-3/x^2} - (1/9) \times \int u e^u du$, putting $(-3/x^2) = u$ so that $(6/x^3)dx = du$

or $z e^{-3/x^2} = (3/8) \times (y+a)^2 e^{-3/x^2} - (1/9) \times (ue^u - e^u) + b$

or $z e^{-3/x^2} = (3/8) \times (y+a)^2 e^{-3/x^2} - (1/9) \times (-3/x^2) e^{-3/x^2} + (1/9) \times e^{-3/x^2} + b$

or $z = (3/8) \times (y+a)^2 + (1/3x^2) + (1/9) + b e^{3/x^2}$, a, b being arbitrary constants.

(b) Proceed exactly as in part (a) **Ans.** $z = (2/3) \times (y+a)^{3/2} + (1/3x^2) + (1/9) + b e^{3/x^2}$

3.22 Ordinary and Partial Differential Equations

Ex. 23. Find the complete integral of $(p + y)^2 + (q + x)^2 = 1$

Sol. Try yourself.

Ans. $z = ax - (1 - a^2)^{1/2} y - xy + b$

Ex. 24. Find the complete integral of $2(y + zq) = q(xp + yq)$ [Nagpur 2003, 06; Delhi B.Sc. (Prog) II 2011, 12; Delhi B.Sc. (Hons) 2011]

Sol. Let $f(x, y, z, p, q) = 2y + 2zq - xpq - yq^2 = 0$... (1)

Charpit's auxiliary equations are $\frac{dp}{f_x + p f_z} = \frac{dq}{f_y + q f_z} = \frac{dz}{-p f_x - q f_y} = \frac{dx}{-f_p} = \frac{dy}{-f_q}$... (2)

$$\frac{dp}{-pq + 2pq} = \frac{dq}{2 - q^2 + 2q^2} = \frac{dz}{2pqx + 2qy - 2qz} = \frac{dx}{qx} = \frac{dy}{xp + 2yq - 2z}, \text{ by (1)}$$

Taking the first and fourth fractions, $(1/pq)dp = (1/qx)dx$ or $(1/p)dp = (1/x)dx$
Integrating, $\log p = \log a + \log x$ or $p = ax$, ... (3)

where a is an arbitrary constant. Substituting the value of p given by (3) in (1), we have

$$2y + 2zq - ax^2q - yq^2 = 0 \quad \text{or} \quad yq^2 + q(ax^2 - 2z) - 2y = 0.$$

$$\Rightarrow q = [-(ax^2 - 2z) \pm \{(ax^2 - 2z)^2 + 8y^2\}^{1/2}] / (2y) \quad \dots (4)$$

Substituting the values of p and q given by (3) and (4) in $dz = p dx + q dy$, we obtain

$$dz = ax dx + (1/2y) \times [2z - ax^2 \pm \{(2z - ax^2)^2 + 8y^2\}^{1/2}] dy$$

or
$$\frac{2dz - 2ax dx}{(2z - ax^2) \pm \{(2z - ax^2)^2 + 8y^2\}^{1/2}} = \frac{dy}{y} \quad \dots (5)$$

Putting $2z - ax^2 = u$ and $2dz - 2ax dx = du$, (5) yields

$$\frac{du}{u \pm (u^2 + 8y^2)^{1/2}} = \frac{dy}{y} \quad \text{or} \quad \frac{du}{dy} = \frac{u}{y} \pm \left\{ \left(\frac{u}{y} \right)^2 + 8 \right\}^{1/2}, \quad \dots (6)$$

which is linear homogeneous differential equation. To solve it, we put $u/y = v$, i.e., $u = yv$ so that $du/dy = v + y(dv/dy)$ and so (6) reduces to

$$v + y \frac{dv}{dy} = v \pm (v^2 + 8)^{1/2} \quad \text{or} \quad \frac{dv}{(v^2 + 8)^{1/2}} = \frac{dy}{y},$$

taking positive sign. Integrating it, we have

$$\log \{v + (v^2 + 8)^{1/2}\} = \log y + \log b \quad \text{or} \quad v + (v^2 + 8)^{1/2} = by$$

or $u/y + \{(u/y)^2 + 8\}^{1/2} = by \quad \text{or} \quad u + (u^2 + 8y^2)^{1/2} = by^2$

or $2z - ax^2 + \{(2z - ax^2)^2 + 8y^2\}^{1/2} = by^2$, as $u = 2z - ax^2$; a, b being arbitrary constants

EXERCISE 3(B)

Using Charpit's method, find a complete integral of the following equations :

- $z = px + qy + pq$. [Mysore 2004] **Ans.** $z = ax + by + ab$
- $pq = xz$. **Ans.** $z = (a + x^2/2)(b + y)$
- $p^2 + px + q = z$. **Ans.** $z = ax + a^2 + be^y$
- $(p + q)(z - px - qy) = 1$. **Ans.** $(a + b)(z - ax - by) = 1$
- $px + qy + pq = 0$ [Meerut 2015] **Ans.** $az = -(1/2) \times (y + ax)^2 + b$
- $q = px + q^2$ **Ans.** $z = (a - a^2) \log x + ay + b$

7. $p - 3x^2 = q^2 - y$ Ans. $z = x^3 - (1/3) \times (a-x)^3 + ay - xy + b$
 8. $x^2p^2 + y^2q^2 = 4$ Ans. $z = a \log x + (4 - a^2)^{1/2} \log y + b$
 9. $xpq + yq^2 = 1$ Ans. $(z+b)^2 = 4(ax+b)$
 10. $p + q = 3pq$ Ans. $az = b - (1/2) \times (y+ax)^2$
 11. $pq + x(2y+1)p + (y^2+y)q - (2y+1)z = 0$ Ans. $z = ax + b(a+y+y^2)$
 12. $z^2(p^2 + q^2) = x^2 + e^{2y}$
 Ans. $\frac{z^2}{2} = \frac{x\sqrt{(x^2+a)}}{2} + \frac{a}{2} \sinh^{-1} \frac{x}{\sqrt{a}} + \sqrt{(e^{2y}-a)} - \sqrt{a} \tan^{-1} \left(\frac{e^{2y}-a}{a} \right) + b$
 13. $p^2 - y^2q = x^2 - y^2$ [Madurai Kamraj 2008]
 Ans. $z = \frac{x\sqrt{(x^2+b)}}{2} + \frac{b}{2} \sinh^{-1} \frac{x}{\sqrt{b}} - \frac{b}{2y^2} + \log y + c$
 14. $p^2q(x^2 + y^2) = p^2 + q$ Ans. $z = \log[x + \sqrt{(x^2+a)}] + \frac{1}{2\sqrt{a}} \log \frac{y-\sqrt{a}}{y+\sqrt{a}} + b$
 15. $yp = 2xy + \log q$. [Lucknow 2010] [Ans. $z = (a+2x)^2 / 4 + (1/a) \times e^{ay} + b$]

3.9 Special methods of solutions applicable to certain standard forms

We now consider equations in which p and q occur other than in the first degree, that is non-linear equations. We have already discussed the general method (*i.e.*, Charpit's method — see Art. 3.7). We now discuss four standard forms to which many equations can be reduced, and for which a complete integral can be obtained by inspection or by other shorter methods.

3.10 Standard Form I. Only p and q present

[Nagpur 2002; Bhopal 2010]

Under this standard form, we consider equations of the form $f(p, q) = 0$ (1)

Charpit's auxiliary equations are $\frac{dp}{f_x + p f_z} = \frac{dq}{f_y + q f_z} = \frac{dz}{-p f_p - q f_q} = \frac{dx}{-f_p} = \frac{dy}{-f_q}$

giving $\frac{dp}{0} = \frac{dq}{0}$, by (1)

Taking the first ratio, $dp = 0$ so that $p = \text{constant} = a$, say ... (2)

Substituting in (1), we get $f(a, q) = 0$, giving $q = \text{constant} = b$, say, ... (3)

where b is such that $f(a, b) = 0$ (4)

Then, $dz = p dx + q dy = adx + bdy$, using (2) and (3).

Integrating, $z = ax + by + c$, ... (5)

where c is an arbitrary constant. (5) together with (4) give the required solution.

Now solving (4) for b , suppose we obtain $b = F(a)$, say.

Putting this value of b in (5), the complete integral of (1) is

$$z = ax + yF(a) + c, \quad \dots (6)$$

which contains two arbitrary constants a and c which are equal to the number of independent variables, namely x and y .

The singular integral of (1) is obtained by eliminating a and c between the complete integral (6) and the equations obtained by differentiating (6) partially w.r.t. a and c ; *i.e.*, between

$$z = ax + yF(a) + c, \quad 0 = x + yF'(a) \quad \text{and} \quad 0 = 1. \quad \dots (7)$$

Since the last equation in (7) is meaningless, we conclude that the equations of standard form I have no singular solution.